

# Drug Decriminalization and College Graduation: Evidence from Oregon's Measure 110

William Swartz

Department of Economics, San Diego State University

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Dr. Penglase

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## **Abstract**

On February 1, 2021, Oregon became the first state in the United States to decriminalize the possession of small quantities of all illicit drugs under Measure 110. This unprecedented policy shift has sparked extensive debate regarding its broader societal implications. Proponents assert that decriminalization can reduce stigma, encourage treatment for substance use disorders, and lessen the criminal justice burden. However, critics caution that such measures could inadvertently increase drug use and overdose rates. This paper constructs a synthetic Oregon, outlined by Abadie, Diamond, and Hainmueller (2010), to examine the causal impact of Measure 110 on college graduation rates in Oregon, offering insights into how drug policy changes may influence educational attainment.

# 1 Introduction

Does drug decriminalization reduce college graduation rates? One can imagine two plausible scenarios in which this policy might influence students' academic outcomes. First, for those who already use drugs, decriminalization—alongside increased social programs addressing substance use—may help them graduate on time by minimizing involvement with the criminal justice system and providing improved treatment options. Alternatively, the policy could raise overall drug use by reducing the legal repercussions of consuming or selling illicit substances, potentially interfering with students' ability to complete their degrees.

Historically, attending college has been considered a protective factor against the development of substance use disorders (Welsh, Shentu, & Sarvey, 2019). However, in a study by Caldeira et al. (2009), nearly half of 946 college students who were followed from freshman to junior year met criteria for at least one substance use disorder during that time. Countless studies have shown that students who regularly use substances are more likely to have lower GPAs, spend fewer hours studying, miss significantly more class time, and fail to graduate on time (Welsh et al., 2019).

College students occupy a unique position in the drug policy debate because they are at a critical transitional stage: many are experiencing newfound independence, encountering academic and social pressures, and establishing long-term habits that can influence their future well-being. This combination of factors makes them both a vulnerable population and a group with significant potential for positive outcomes if appropriate support structures and policies are in place.

Moreover, college graduates typically have higher lifetime earnings, lower unemployment rates, and are more likely to engage civically. Ensuring that students can remain enrolled and progress toward graduation—rather than dropping out due to legal issues, health problems, or financial strains associated with substance use—directly contributes to the economic and social welfare of the state. Consequently, when contemplating drug policy reforms, the potential impact on college students' academic success and subsequent professional oppor-

tunities becomes a vital consideration, as it links public health interventions to broader educational and economic goals.

Despite the importance of understanding how statewide drug policies affect educational outcomes, there appears to be no existing research examining Measure 110’s impact on college graduation rates. This paper addresses that gap by providing the first causal estimates of how decriminalization under Measure 110 has influenced college completion in Oregon.

## 2 Institutional Background

Historically, U.S. drug policy has been largely rooted in prohibition and strict enforcement, highlighted by the “War on Drugs” launched in the 1970s by President Richard Nixon, resulting in widespread criminalization of drug-related activities and a surge in incarceration rates, often disproportionately affecting minority groups. Throughout the 1980s and 1990s, mandatory minimum sentencing laws reinforced this punitive model, raising concerns about its social and economic costs. By the late 20th and early 21st centuries, policymakers and health professionals increasingly questioned the efficacy of a purely punitive approach, pointing to issues of addiction, public health, and prison overcrowding. In response, some regions began experimenting with alternative frameworks—most notably Portugal in 2001, which decriminalized all illicit drugs and channeled resources toward treatment and harm reduction. Within the United States, Oregon’s Measure 110 represents the first comprehensive statewide effort to decriminalize the possession of all drugs for personal use. Reflecting a broader shift toward viewing addiction as a public health challenge rather than a criminal one, such measures aim to reduce stigma, encourage treatment, and ultimately address the root causes of substance abuse.

## 2.1 Measure 110

Measure 110, passed in November 2020, was a law aimed at shifting the state’s approach to substance use from criminal punishment to health-oriented treatment. Taking effect on February 1, 2021, it reduced penalties for possessing small amounts of certain controlled substances—such as cocaine, heroin, methamphetamine, and ecstasy—from felonies or misdemeanors to a new Class E violation, carrying a maximum fine of \$100. Prior to this, possession of these substances was a Class A criminal misdemeanor punishable in Oregon by up to a year in prison and/or fines up to \$6,250. Alongside this decriminalization, the law established Behavioral Health Resource Networks (BHRNs) aimed at providing treatment and support, funded by redirected marijuana tax revenue and law enforcement savings.<sup>1</sup> However, in 2024, drug possession was recriminalized following a campaign by opponents of the original bill who argued the measure had failed to address the drug crisis effectively. The Oregon legislature passed House Bill 4002 in 2024, rolling back parts of Measure 110.<sup>2</sup>

## 3 Literature Review

One of the primary concerns about changing drug policies is their potential impact on youth education and development, especially given the substantial body of public health and health economics research linking drug use to negative academic and longer-term life outcomes. Chatterji (2003) explores the relationship between adolescent drug use and educational attainment, finding a negative association between marijuana and cocaine use during high school and the number of years of schooling completed. Using state-level drug policies as instruments, Chatterji’s findings suggest that substance use during critical educational years has a significant detrimental effect on long-term educational outcomes, reducing educational attainment by 0.2 to 0.4 years for those who used marijuana or cocaine during high school

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<sup>1</sup><https://olis.oregonlegislature.gov/liz/2019I1/Downloads/CommitteeMeetingDocument/227319>

<sup>2</sup><https://www.opb.org/article/2024/08/29/measure-110-drug-law-deflection-possession-crime-law-oregon-recriminalization-decriminalization/>

(Chatterji, 2003). These findings are further reinforced by Arria, Caldeira, Bugbee, Vincent, and O’Grady (2015), which find, via a longitudinal cohort study and structural model, that marijuana use in college students is associated with various negative educational outcomes including more missed classes, lower GPA, and longer times to graduation. Similarly, Mezza and Buchinsky (2021) develop a dynamic structural model that integrates decisions on drug use, education, and labor force participation. Their findings indicate that early drug use, particularly before age 19, is associated with lower educational attainment and poorer labor market outcomes later in life. This study emphasizes the potential long-term ramifications of youth exposure to drugs, suggesting that increased access to substances due to decriminalization could hinder educational achievement by altering incentives to invest in schooling (Mezza & Buchinsky, 2021).

Health outcomes, particularly overdose rates, are a critical area of concern when evaluating the impact of drug decriminalization. Spencer (2023) examines the causal effect of Oregon’s Measure 110 on drug overdose deaths using the synthetic control method. He finds that decriminalization led to a 23% increase in drug overdose deaths in Oregon during the first year post-implementation, representing 181 additional fatalities compared to a synthetic control group constructed from other U.S. states (Spencer, 2023). Contrasting these findings, Zoorob, Park, Kral, Lambdin, and Del Pozo (2024) present a more nuanced analysis that controls for the rapid introduction of fentanyl into Oregon’s unregulated drug market. They find that the surge in fatal overdoses in Oregon coincided with the influx of fentanyl, making it difficult to attribute the increase solely to decriminalization. After accounting for fentanyl as a potential confounder, the effect of Measure 110 on overdose mortality becomes statistically insignificant (Zoorob et al., 2024). This suggests that, while the timing of Measure 110’s implementation and increased overdoses were correlated, fentanyl’s emergence may have been the primary driver, though whether Oregon had a more drastic increase in fentanyl supply compared to other states is unclear.

Because Measure 110 focused on harm reduction rather than punitive measures, it is

crucial to situate this policy within the broader context of similar approaches. Maghsoudi, McDonald, and Thompson (2021) conduct a systematic review of drug checking services, which aim to provide chemical analysis of drugs to reduce harm among users. While these services have been primarily used in European contexts, findings indicate that providing users with accurate information about drug composition can alter consumption patterns and reduce risky behaviors, such as the use of contaminated substances (Maghsoudi et al., 2021). This suggests that complementary harm reduction strategies may be essential to mitigate the risks of decriminalization.

Although the literature consistently points to the negative effects of drug use and remains inconclusive regarding Measure 110’s impact on overdoses, the key question is how a more lenient drug policy might influence drug use among youth and college-age populations. In Oregon specifically, Kerr, Bae, and Koval (2018) find that recreational marijuana laws (RMLs) did not significantly increase marijuana use among college students, though it is important to note that marijuana use of college-aged students may be endogenous to RMLs being passed. Wen, Hockenberry, and Cummings (2015), on the other hand, examine the effects of Medical Marijuana Laws (MMLs) on various substances. They find that while marijuana use increased among adults post-MML, there were no significant changes in alcohol or other drug use among youth, indicating that perceptions and enforcement play a crucial role in shaping policy outcomes (Wen et al., 2015).

Despite the existing body of research on drug decriminalization and its health and behavioral impacts, there is a gap in understanding how policies regarding ‘harder’ drugs may affect educational outcomes, particularly for college students. Given the significant controversies regarding Measure 110 and the potential for increased drug exposure in communities post-decriminalization, it is essential to explore whether these changes have created an environment that is conducive to college completion. By focusing on college graduation rates, this study aims to fill a critical gap in the literature and provide insights into how policy shifts in drug regulation may shape the educational trajectories of youth and young adults.

Understanding these dynamics is crucial for policymakers as they navigate the complex balance between public health, safety, and educational priorities in the context of drug policy reform.

## 4 Data

I use data from the National Center for Education Statistics (NCES), specifically the Integrated Postsecondary Education Data System (IPEDS), along with the American Community Survey (ACS) 1-year tables. IPEDS is a system of 12 interrelated survey components conducted annually that gathers data from every college, university, and technical and vocational institution that participates in the federal student financial aid programs. The data collections occur in fall, winter, and spring. This is the most comprehensive postsecondary, college-inclusive data collection available. Within the IPEDS survey, I gather institutional characteristic data, graduation rates, and enrollment figures at the institution level from 2012 to 2023. I then merge these with state-level economic data from the ACS for the same period, creating a master dataset from which I construct a synthetic Oregon.

Table 1: Descriptive Statistics

| Variable         | Obs | Mean      | Std. Dev. | Min       | Max        |
|------------------|-----|-----------|-----------|-----------|------------|
| Graduation Rate  | 612 | 47.442    | 7.555     | 24.061    | 70.500     |
| Enrollment Total | 561 | 4546.794  | 1866.839  | 1883.741  | 16054.120  |
| Grad Degree      | 561 | 61.489    | 36.403    | 8.300     | 93.900     |
| Unemp. Rate      | 561 | 5.665     | 1.962     | 2.200     | 12.300     |
| Median Income    | 561 | 62108.014 | 13282.897 | 37095.000 | 108210.000 |

Graduation Rate and Enrollment Total come from IPEDS. Enrollment Total was missing for 2023 since it had yet to be calculated. Grad Degree comes from the ACS, which is the estimated percent of civilian population 25 years or older with a professional or graduate degree. The remaining variables also come from the ACS. All ACS estimates were missing for 2020. No missing predictor variables are of concern when constructing synthetic Oregon since no pre-treatment data is missing.

I restrict my sample to institutions offering two-year or higher degrees. Table 1 shows pooled averages of my variable of interest, college graduation rates, as well as the predictors used to construct synthetic Oregon. Total enrollment is collected yearly at the institutional

level, which I then collapse by year and state; thus the table shows the average total enrollment by institution by state.

Figure 1 shows the country-wide average graduation rate. The graduation rate is calculated by IPEDS by cohort—full-time, first-degree-seeking students in a particular year—that have completed their program within 150% of the normal time period. For a 4-year degree this would mean that the student completes their degree within 6 years of entering the program. For a specific year, say 2018, this is the cohort graduating in the spring of 2019. As seen in Figure 1, the country-wide graduation rate, for all institutions, has increased by about 5 percentage points. Even though Measure 110 was passed in November of 2020 and did not go into effect until February 2021, this is still our ‘2020’ cohort.

Figure 2 shows Oregon’s average graduation rate over time versus the rest of the country. Figures 3 and 4 split Figure 2 by public and private institutions. In Oregon the graduation rate for public institutions is lower on average than the rest of the country but is trending upwards along with the country, while private institutions in Oregon have higher graduation rates on average than the rest of the country but are much more volatile and seem to be decreasing. This difference motivates opportunities for later heterogeneity analysis. All figures show averages for both 2- and 4-year institutions.

## 5 Econometric Strategy

In this paper, I estimate the effect of Measure 110 on college graduation rates using a synthetic control analysis, a method well-suited to policies enacted in a single state. This approach involves constructing a “synthetic” Oregon from a donor pool consisting of the other 49 states and Washington, D.C., thereby allowing me to measure the Average Treatment Effect on the Treated (ATT)—that is, the difference in graduation rates with and without Measure 110. To do this, I select a set of predictors to weight the donor pool, minimizing the pre-treatment Mean Squared Prediction Error (MSPE) and ensuring an optimal



Figure 1: Country-Wide Average Graduation Rate

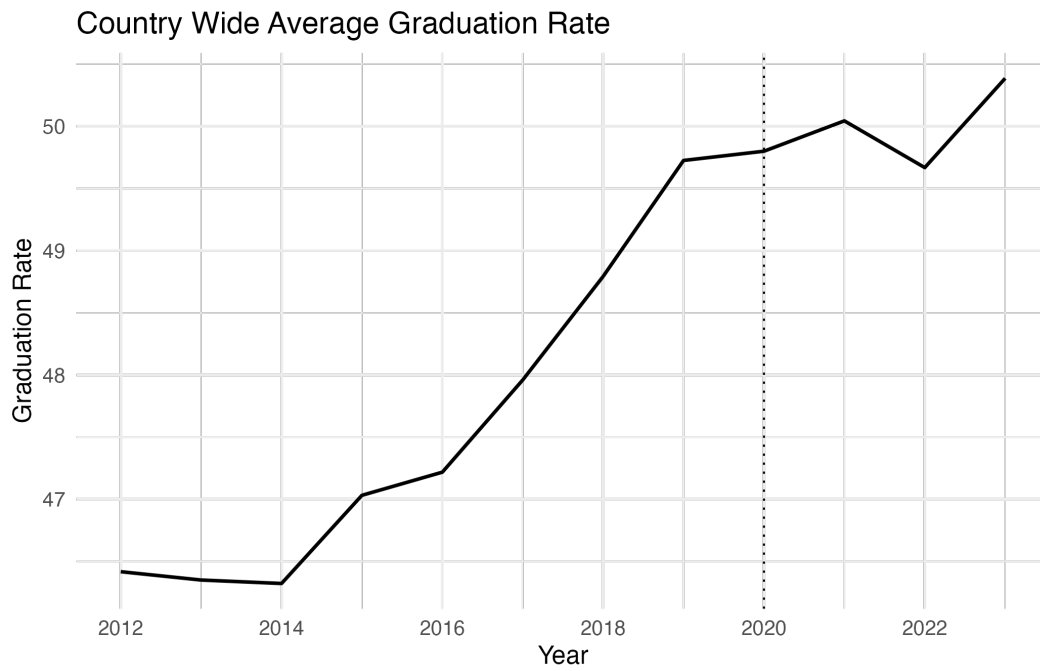


Figure 2: Average Graduation Rate, Oregon vs Other States

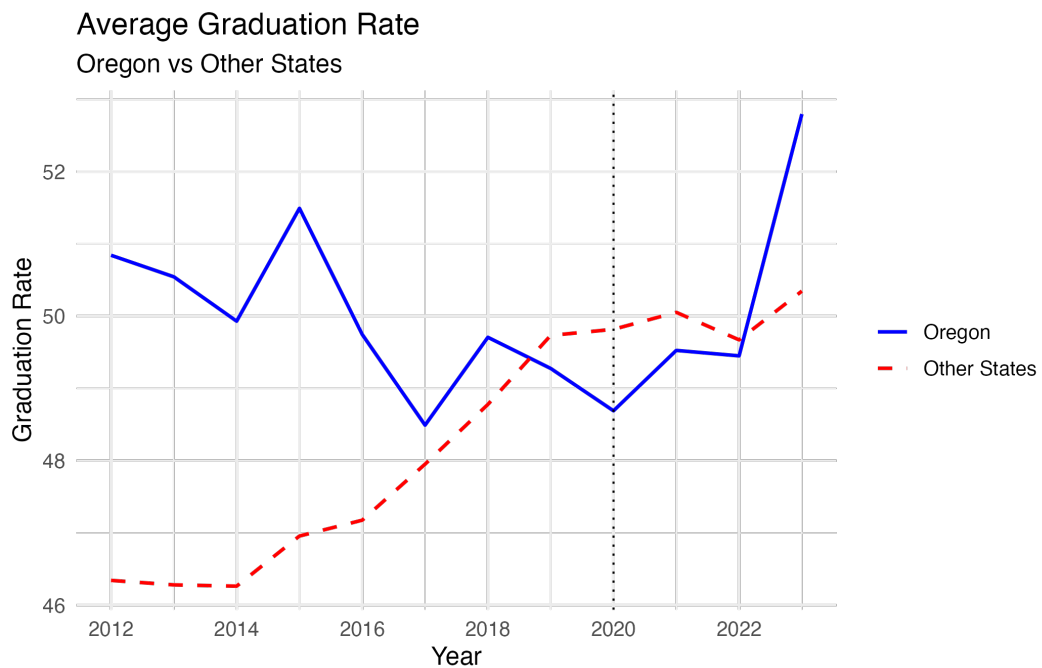


Figure 3: Average Graduation Rate in Public Universities

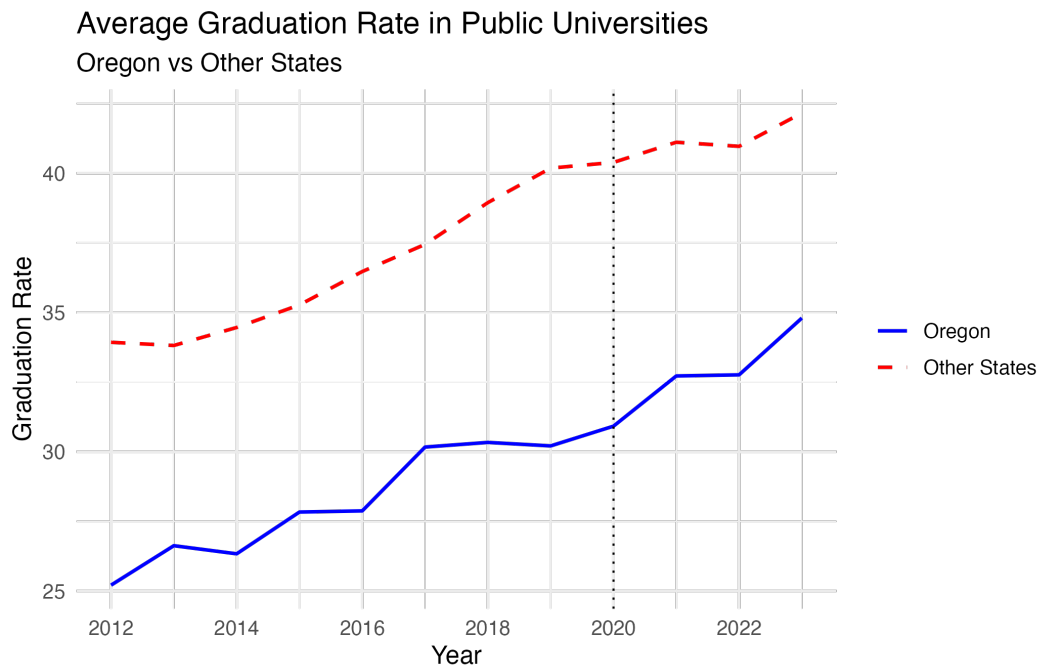
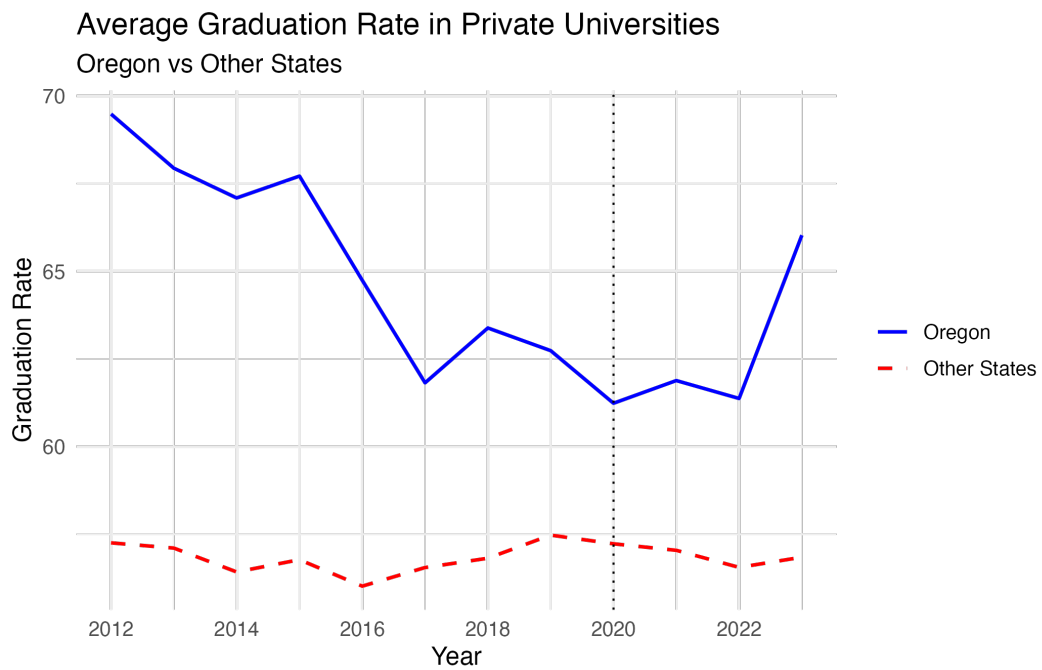


Figure 4: Average Graduation Rate in Private Universities



counterfactual.

Let  $Y_{\text{Oregon},t}$  be the observed outcome, the college graduation rate in Oregon at time  $t$ , and let  $Y_{\text{synthetic},t}$  be the outcome for the “synthetic Oregon” at the same time  $t$ . The synthetic unit is a weighted average of the other states (the donor pool), with weights chosen to match Oregon’s pre-treatment characteristics. Denoting the set of donor states as  $j = 2, 3, \dots, J$  and the corresponding optimal weights  $w_j^*$ , we can write:

$$Y_{\text{synthetic},t} = \sum_{j=2}^J w_j^* Y_{j,t} \quad (1)$$

Thus,

$$\text{ATT}_t = Y_{\text{Oregon},t} - Y_{\text{synthetic},t}. \quad (2)$$

Despite the advantages of employing a synthetic control approach, several challenges may arise in this setting. The effects of decriminalization on educational outcomes may take several years to manifest; a short, in this case three-year, post-treatment window might not fully capture Measure 110’s long-term impact. Concurrent policy changes—such as alterations in financial aid, tuition costs, or other statewide reforms—can further complicate the isolation of Measure 110’s effect, though I did not find any evidence of this in my research. Finally, constructing a valid synthetic counterfactual requires identifying a donor pool of states that credibly resembles Oregon’s pre-treatment trends, which can be difficult if Oregon is an outlier or if no suitable combination of donor states exists—a violation of the convex hull assumption.

One of the most significant threats to identification is the effect of COVID-19 on graduation rates, which coincided with the implementation of Measure 110. Although the pandemic affected every state, local responses, severity, and the resulting impact on educational outcomes likely varied. Ideally, controlling for a time fixed effect and including an interaction term between COVID-19 severity and state would help mitigate these confounding effects. I believe a Synthetic DiD framework may better suit this kind of analysis in further research.

To credibly estimate a causal impact, several assumptions must hold. Most critically, the synthetic control method relies on the assumption that, in the absence of treatment, Oregon would have followed the same path as the composite of donor states. This requirement necessitates strong pre-treatment fit between Oregon and the synthetic counterpart, which is explored in my preliminary results. Another key assumption is that Measure 110 does not significantly alter outcomes in other states, since potential cross-border effects would compromise the donor states' validity. However, no border states are given any weight in my donor pool. Ideally, I would like to have removed individual institutions that border Oregon but due to time constraints, this was not possible. I am not worried about student migrations due to IPEDS' calculation of graduation rates being via cohort and controlling for transfers in and out. Though even if you did believe graduation rates were being affected by student migration because of Measure 110, this is still a valid effect within the ATT framework.

Additionally, since I have data at the institution level and we see that graduation rates differ across institution sectors, some heterogeneity analysis is warranted. To do this, I subsample across sector and level, meaning public vs private and urban vs non-urban institutions. Splitting the sample allows the treatment effects to be different across college types.

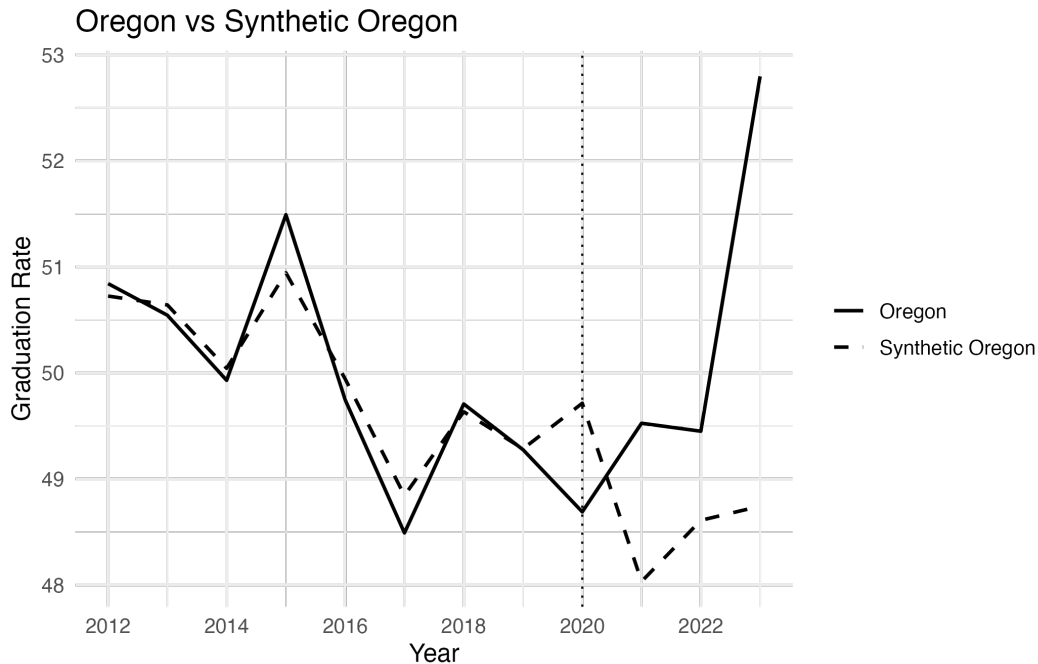
A final important piece of context in this analysis is that Measure 110 may have influenced graduation rates primarily through its impact on drug use, making any observed effect effectively a secondary or indirect policy outcome. Ideally, I would have directly measured changes in drug use following Measure 110 and then employed mediation analysis to determine the extent to which drug use explained shifts in graduation rates. Unfortunately, time constraints prevented me from conducting that additional layer of analysis.

## 6 Preliminary Results

In this section, I present my preliminary results and discuss their implications. I begin by reporting the main findings without subsampling the data and then explore potential heterogeneous treatment effects by splitting the dataset based on various institutional characteristics.

Figure 5 shows the synthetic control graph, comparing Oregon’s graduation rates to those of its synthetic counterpart over time. A visual inspection reveals a strong pre-treatment fit, consistent across all subsequent subsamples and specifications. While a naïve interpretation might suggest that Measure 110 increased Oregon’s graduation rate by 4 percentage points by 2023, in-space placebo tests do not support this conclusion. As shown in Table 2, the p-values for all estimates are insignificant, and Figure 6 further demonstrates that this outcome is not unique.

Figure 5: Oregon vs Synthetic Oregon



For the rest of this section I show just the tables, like Table 2, for the remaining subsamples, none of which result in any significant effects.

Figure 6: Placebo Tests of Oregon vs Other States

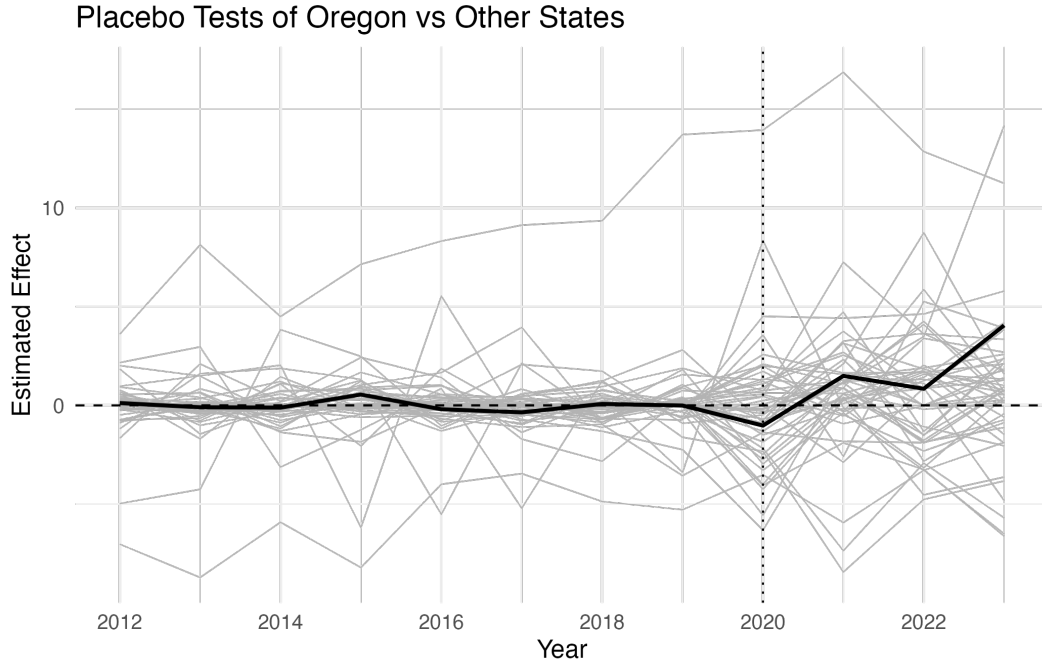


Table 2: ATTs and Associated P-Values, All Institutions

|      | Estimate | P-value | Standardized P-value |
|------|----------|---------|----------------------|
| 2020 | -1.024   | 0.58    | 0.420                |
| 2021 | 1.492    | 0.44    | 0.320                |
| 2022 | 0.840    | 0.86    | 0.520                |
| 2023 | 4.053    | 0.16    | 0.160                |

All institutions. ATTs and p-values constructed via in-space placebo tests (R implementation of `synth_runner`).

Table 3: ATTs and Associated P-Values, Public Institutions

|      | Estimate | P-value | Standardized P-value |
|------|----------|---------|----------------------|
| 2020 | -0.149   | 0.94    | 0.940                |
| 2021 | 1.136    | 0.60    | 0.460                |
| 2022 | 0.083    | 0.92    | 0.880                |
| 2023 | 3.001    | 0.26    | 0.320                |

Public institutions. ATTs and p-values constructed via in-space placebo tests (R implementation of `synth_runner`).

Table 4: ATTs and Associated P-Values, Private Institutions

|      | Estimate | P-value | Standardized P-value |
|------|----------|---------|----------------------|
| 2020 | -1.449   | 0.72    | 0.620                |
| 2021 | 1.741    | 0.64    | 0.620                |
| 2022 | 1.595    | 0.60    | 0.560                |
| 2023 | 7.709    | 0.10    | 0.140                |

Private institutions. ATTs and p-values constructed via in-space placebo tests (R implementation of `synth_runner`).

Table 5: ATTs and Associated P-Values, Urban Institutions

|      | Estimate | P-value | Standardized P-value |
|------|----------|---------|----------------------|
| 2020 | -0.987   | 0.84    | 0.800                |
| 2021 | -1.742   | 0.58    | 0.620                |
| 2022 | -1.398   | 0.68    | 0.640                |
| 2023 | 3.870    | 0.44    | 0.520                |

Urban institutions. ATTs and p-values constructed via in-space placebo tests (R implementation of `synth_runner`).

## 7 Conclusion

I did not find any evidence that Measure 110 influenced college graduation rates under any of the specifications or subsamples tested. Future research may be warranted to first determine whether Measure 110 affects college students’ drug use, as that could provide a clearer link between the policy and educational outcomes. However, based solely on the results here, there is no indication that Measure 110 has a significant impact on graduation rates. Exploring further subsampling and heterogeneity analyses across different demographic groups and institutional characteristics could be a promising direction for additional study.

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